

Project Description

Background

A 2021 study found that 63.0% of surveyed older adults stated driving was their primary mode of community mobility and transportation (Marfeo et al., 2021). Nationally, approximately 57 million individuals live with disabilities, and over 40% rely on others for transportation due to personal and environmental barriers that prevent them from driving (Dicianno et al., 2021). However, only 9.2% reported receiving rehabilitation services focused on driving or transportation-related goals in the past year (Marfeo et al., 2021). These barriers—including limited access to adapted vehicles, decreased driving confidence, and lack of knowledge about driving resources—can lead to social isolation, reduced access to healthcare, and diminished quality of life (Park et al., 2023). Driving and community mobility are among the most important meaningful occupations individuals can engage in, facilitating performance in other areas of life including leisure, work, and social participation (Stav, 2012).

As older adults age, barriers to community mobility can be addressed through appropriate specialized driving rehabilitation services. Despite this service option, the Co-PI is currently the **only** Driving Rehabilitation Professional and NYS Certified Driving Instructor on Long Island who has a healthcare background and education, creating an immense opportunity for the study team to meet regional driving rehabilitation needs. Traditional driving rehabilitation services involve comprehensive clinical evaluations of vision, cognition, and physical/motor skills, and on the road assessments and training. However, evidence suggests that naturalistic driving methods deploying mobile health technology may be engaging and show greater representativeness of performance than traditional driving assessments for older adults (Babulal et al., 2016; Bayat et al., 2021; Cammarata, 2024).

In fact, in-situ behavioral data collection via sensors and passenger real-time inputs may have great clinical value and provide a richer context of driving behaviors to evaluate older adults' driving safety and performance. When combined with self-reports and passenger rated performance, sensor-based application will provide a more comprehensive picture of driving rehabilitation intervention needs in a more naturalistic experience. Importantly, including passengers—often caregivers or family members—integrates a novel opportunity with observing the driver's behaviors, safety, and concerns following the prompted in-app system that is categorized by driving skills. Mobile health interventions offer opportunities to individualize targeted training programs and deliver goal-specific driving supports in situ as an extension of 1:1 clinical rehabilitation sessions. The smartphone presents a ubiquitous vehicle to collect objective sensor data and subjective performance (report) data and, importantly, presents the potential for a performance feedback system targeted to the individual drivers.

The proposed project recognizes the variability in driver needs and supports dynamic customization of its interventions based on current functional abilities. Via the proposed mobile health methods, educational content is deployed in a scaffolded manner as the driver demonstrates consistency and safety with skills outside of clinical sessions. This approach enables just-in-time interventions based on caregiver data inputs, clinician alerts, or goal-based strategies, and ultimately could enhance validity of performance data that supports individualized, continuous care and aligns with both the client's goals and the rehabilitation plan. Together, caregiver support and the mobile health approach bridges key elements to create the foundation for advancing innovative practice. The proposed study utilizes technology to extend traditional rehabilitation services and promote an individualized approach that can also serve as a tool for caregivers, family members, and driver rehabilitation professionals. The overall goals of this project are to elevate the standards of traditional assessments by developing actionable tools and enhance driving safety during real-world driving experiences.

Specific Aims

Driving and community mobility are critical for older adults' independence, health access, and quality of life. However, older adults and individuals with disabilities face personal and environmental barriers to safe driving, and very few receive rehabilitation services due to the lack of services or availability. There is a critical need for accessible, interdisciplinary driver rehabilitation programs and data-driven tools to guide clinical decisions. This project aims to fill this gap and test a novel, mobile health app during

real-world experiences while integrating clinical evaluations and passenger feedback to support clinical decision-making.

Aim 1: Evaluate the validity of the mobile-health app with assessing the domains across Michon’s driving behavior levels by comparing them to established cognitive, visual, and motor assessments.

App-based measures will correlate with standard clinical tests, demonstrating concurrent validity.

Aim 2: Evaluate the feasibility and validity of a mobile health app platform for collecting real-world data on driver behavior and self-reported fitness to drive among older adults.

Aim 3: Explore the relationship between clinical evaluation results and real-world driver feedback collected via the smartphone-based platform. This aim will examine the alignment of clinical assessments with feedback and trip patterns reported by drivers and caregivers. Insights from this comparison will help guide future, personalized approaches to driver rehabilitation and safety training.

Methodical Approach and Timeline

The investigators have laid the groundwork for this study to be completed. Community stakeholders and the target population have been identified as part of a projected, August 2025, pilot program, essential for tailoring the mobile health app. Educational materials and workshops have been developed to highlight the benefits of driver rehabilitation, screening procedures, and the safety impact of these services. These tools will also support program expansion beyond the SBM network. Additionally, following the Co-PI’s invited presentation at the SBM Town Hall, Drs. Verghese and Ambrose have both committed to participating in this study by supporting the developed provider referral and screening process and pre-screening participants for this project. These steps have positioned the PI’s team to expand on intervention and clinical assessment tools to develop an effective Driver Rehabilitation Program.

Study Eligibility. Participants will be recruited through a physician referral process developed by the Co-PI team and with Drs. Verghese and Ambrose. Eligible participants include fully-licensed adult drivers (Class D–NYS) accompanied by an adult passenger who also holds a valid license. Both must own smartphones capable of running the study platform.

Smartphone-based assessments will capture driving behavior in participants’ natural environments, thus minimizing recall bias and enhancing ecological validity. To address Aim 1, drivers and caregivers will use the platform to provide brief post-trip reflections: drivers will rate performance and confidence, and passengers will log safety behaviors and comfort levels using a checklist generated from that week’s intervention focus. Study staff will support users and establish personalized prompts based on participant goals and in-clinic performance. We will collect basic in-car or pre-operational skills and also deliver objective, real time feedback on driver safety following Michon’s Hierarchical Model of Driving Behavior which categorizes driving skills into 3 levels (strategic, tactical, and operational).

We will explore correlations between clinical assessment scores and smartphone-derived data. This includes comparing objective clinical findings (vision, cognition, motor) with driver self-reports and passenger safety observations to address Aim 2. These analyses will generate early evidence for aligning clinical indicators with real-world behavior, strengthening the case for scalability, personalized driving rehabilitation interventions, and informing future funding proposals. Finally, we will analyze the relationships between clinical findings, self-reported driver performance, and caregiver safety observations to address Aim 3. The resulting data will help establish early evidence linking in-clinic assessments with real-world behaviors. These findings will inform future program development, support scalability, and contribute to personalized driving rehabilitation planning.

Project Timeline

Month	Description
August 2025	School of Health Professions (SHP) Seed Grant Submission; Finalize physician referral procedures

September 2025	Grant Awarded; purchase projects budget items; Update and refine smartphone platform; IRB proposal submission
October 2025	Internal auditing of the smartphone platform and clinical evaluation procedures
November 2025	Finalize policies and procedures for the clinic;
January 2026	Begin community outreach and education
February 2026	Begin recruitment following physician referred procedures; Platform testing continues Draft update report to the SHP
March 2026	Begin clinical assessments and participant onboarding to smartphone platform; Data collection; Submit update report to the SHP
April 2026	Prepare external grant LOI (Mother Cabrini Health Foundation)
May 2026	Final data collected; Data analysis
June 2026	Findings confirmation; ASHAP Interprofessional Collaboration Research Grant Submission
July 2026	Draft manuscript and/or conference submission and the SHP Seed grant report
August 2026	Begin Grant LOI Submission (AOTF Implementation/Intervention); Submit final SHP Seed grant report

Potential Study Impact

Suffolk County has a significant population of older adults (nearly 20% aged 65 and older) and individuals with disabilities (approximately 8% under the age of 65) (U.S. Census Bureau, 2023). These barriers can impact social engagement, independence, and a sense of identity (Dickerson et al., 2024). New York State Office For the Aging conducted a statewide needs assessment survey for older adults from February 2023 through May 2023 (NYSOFA, 2025). This data identifies challenges that New York drivers are experiencing and with limited support. Driving cessation is also a risk for older drivers and drivers with disabilities and who often resist it due to the decreased activity engagement and independence (Dickerson, 2024).

Driving rehabilitation programs do not currently implement smartphone based technology or utilize passenger input for assessment and responsive intervention as naturalistic driving methods. Our approach to explore these methods and contrast them with traditional tools positions our team at the cutting edge of driving rehabilitation research. This approach also has great potential impact for clinical care, introducing only nominal additional expenses (data storage, platform servicing) because of the ubiquity of smartphones, and presents the opportunity to extend clinical services that potentially lead to more effective driving rehabilitation outcomes. Furthermore, this project directly addresses a critical service gap on Long Island, where a growing population of older adults and individuals with disabilities face transportation barriers that affect independence and well-being. This project will build on a pilot program taking place August 2025 and create a sustainable driving rehabilitation program by testing real-world behavior monitoring. This project will:

- Validate the mobile health app by comparing app-based measures of driving behavior (strategic, tactical, operational) with established clinical assessments in vision, cognition, and motor function (Aim 1)

- Evaluate the feasibility and utility of a smartphone-based platform to collect real-world data on driving behavior and self-reported fitness to drive among older adults (Aim 2)
- Explore the relationship between clinical evaluation results and real-world driver and caregiver feedback to support personalized, community-based driving interventions (Aim 3)

Overall, the pilot findings from our proposed SHP Seed Grant will support these innovative research and scholarship activities that position the PI team for future foundation funding opportunities. This project provides a foundation for publications, research collaborations, and interprofessional studies related to driving rehabilitation. The pilot program's data will be used to validate driving assessments and intervention strategies, making the project highly competitive for national grants. The investigators plan to apply for additional funding, including the American Occupational Therapy Foundation's Intervention Research and Implementation Research Grants (December 2025) the Mother Cabrini Health Foundation (June 2026), and ASAHP's Interprofessional Collaboration Research Grant (June 2026).

This project aligns with the SHP Research Seed Grant's goals to support **clinical advancement**, **healthcare education**, and **technological innovation**, and has strong potential to attract larger external funding, including foundation grants (e.g., AOTF, ASAHP, Mother Cabrini Health Foundation). This project will also have a strong interdisciplinary collaboration between the School of Health Professions' occupational therapy program and Stony Brook Medicine's Neurology Department. Both Drs. Ambrose and Verghese have begun this collaboration with the PI team by exploring the SBM physician referral system and recommend participants for this project by participating in refining the physician internal to SBM referral procedures, as well as, screen and recommend patients for this study to Finally, the overall program will reduce occupational disparities and inform the community and health care providers on a community resource to support safe, independent mobility for aging and disabled populations.

Roles and responsibilities of PI and Co Investigators

The proposed study builds on the PI's (Dr. Hannah Mercier) clinical research expertise in technology-based methods to assess and promote health, mobility, and community participation among adults with disabilities. Dr. Mercier will lead the study design, data analysis, and integration of the mobile platform within the program. Acquiring SHP Seed Funding will strengthen her research portfolio and position her as a competitive applicant for future federal and foundation grants in digital health and rehabilitation.

The Driver Rehabilitation Program will be led by Dr. Rosalia Casanova, a licensed occupational therapist, certified Driving Rehabilitation Professional (DRP), and certified Driving Instructor (CDI) in New York State. The Co- PI will oversee the clinical aspects of the program, including protocol development, administration of standardized assessments, and participant screening. Their unique qualifications enable them to translate clinical evaluation findings into real-world driver recommendations, bridging research and practice. Together, the PI and Co-PI bring complementary strengths in clinical research, technology integration, and driver rehabilitation, ensuring the project is methodologically sound, clinically relevant, and positioned for scalability through future external funding.

Plans for Future Grant Submission and Dissemination

April 2026	Grant LOI submission (Mother Cabrini Health Foundation)
June 2026	ASAHP Interprofessional Collaboration Research Grant Submission
August 2026	Grant LOI Submission (AOTF Implementation Grant)

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